

EECE 571Q

Lecture 06: The Clifford group

Thursday 28 May 2026

Announcements

- Finalizing project groups
- Quiz 3 on Tuesday; back in person

Relationship with stabilizer codes

We defined binary symplectic representation as an isomorphism between $\hat{\mathcal{P}}_n$ and vectors in $\mathbb{Z}_2^n \times \mathbb{Z}_2^n$.

$$I = (0 \mid 0) \quad X = (1 \mid 0) \quad Z = (0 \mid 1) \quad Y = (1 \mid 1)$$

$$Z \otimes Y \otimes I \otimes X = (0 \ 1 \ 0 \ 1 \mid 1 \ 1 \ 0 \ 0)$$

Commutation:

$$P_1 = (x_1 \mid z_1) \quad P_2 = (x_2 \mid z_2)$$

$$\begin{array}{r} XYZ \\ ZYZ \end{array} \quad \begin{array}{l} (1 \ 1 \ 0 \mid 0 \ 1 \ 1) \\ (0 \ 1 \ 0 \mid 1 \ 1 \ 1) \end{array}$$

$$V_1 \odot V_2 = x_1 \cdot z_2 + z_1 \cdot x_2 \pmod{2}$$

$$\begin{aligned} &= (1 \ 1 \ 0) \cdot (1 \ 1 \ 1) + (0 \ 1 \ 1) \cdot (0 \ 1 \ 0) \\ &= 0 + 1 = 1 \end{aligned}$$

1: anti commute
0: commute

$\underbrace{X \checkmark \checkmark}_{\text{odd}} \rightarrow \text{anticommute}$

Last time

We made an association with classical codes and stabilizers, and defined **CSS codes**.

(Source: Gottesman Thm. 5.1)

Let C_1 be an $[n, k_1, d_1]$ classical linear code with parity check matrix H_1 , and C_2 be an $[n, k_2, d_2]$ classical linear code with parity check matrix H_2 , with $C_1^\perp \subseteq C_2$. The stabilizer code (CSS code) with stabilizer generators

$$\left(\begin{array}{c|c} 0 & H_1 \\ H_2 & 0 \end{array} \right)$$

is an $[[n, k, d]]$ quantum code with $k = \underbrace{k_1 + k_2}_{\text{number of rows}} - n$ and $d \geq \min(d_1, d_2)$.

Learning outcomes

- ➊ Derive parameters for the smallest code that corrects arbitrary single-qubit errors (and express it in the stabilizer formalism)
- ➋ Define the Clifford group and compute its action on stabilizer generators
- ➌ Perform efficient classical simulation of Clifford circuits

The Hamming bound

What is the smallest (non-degenerate) code that corrects arbitrary single-qubit errors?

Shor: $[[9, 1, 3]]$

Steane: $[[7, 1, 3]]$

General bound:

- n physical qubits
- k logical qubits
- up to t errors (X, Y, Z)

$$\underbrace{0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ \dots}_n \quad \dim(\mathcal{H}_n) = 2^n$$

If a non-degenerate code exists,

$$\text{how many different errors?} \quad \textcircled{X} 2^k \leq 2^n$$

$$0 \text{ error: } 1 = \binom{n}{0} \cdot 3^0$$

$$1 \text{ error: } \binom{n}{1} \cdot 3$$

$$2 \text{ errors: } \binom{n}{2} \cdot 9 = \binom{n}{2} \cdot 3^2$$

$$t \text{ error: } \binom{n}{t} \cdot 3^t$$

$$\left(\sum_{j=0}^t \binom{n}{j} 3^j \right) \cdot 2^k \leq 2^n$$

$$k=1, t=1 \Rightarrow (1+3n) \cdot 2 \leq 2^n \Rightarrow \boxed{n=5}$$

The smallest code: $[[5, 1, 3]]$

$$|0_L\rangle = \frac{1}{4} \left[|00000\rangle + |10010\rangle + |01001\rangle + |10100\rangle \right. \\ \left. + |01010\rangle - |11011\rangle - |00110\rangle - |11000\rangle \right. \\ \left. - |11101\rangle - |00011\rangle - |11110\rangle - |01111\rangle \right. \\ \left. - |10001\rangle - |01100\rangle - |10111\rangle + |00101\rangle \right]$$

$$|1_L\rangle = \frac{1}{4} \left[|11111\rangle + |01101\rangle + |10110\rangle + |01011\rangle \right. \\ \left. + |10101\rangle - |00100\rangle - |11001\rangle - |00111\rangle \right. \\ \left. - |00010\rangle - |11100\rangle - |00001\rangle - |10000\rangle \right. \\ \left. - |01110\rangle - |10011\rangle - |01000\rangle + |11010\rangle \right]$$

Name	Operator
g_1	$XZZXI$
g_2	$IXZZX$
g_3	$XIXZZ$
g_4	$ZXIXZ$
$\bar{Z}$	$ZZZZZ$
$\bar{X}$	$XXXXX$

Screenshot: Nielsen & Chuang, chapter 10.5.6.

The quantum Gilbert-Varshamov bound

Quantum Hamming bound tells us parameters of smallest code.
What about existence criteria for *any* code?

(Source: Gottesman Thm. 7.1) $[[n, k, d]]$

size of logical space (pointing to k)
logical qubits (pointing to k)

There exists an $((n, 2^k, d))$ QECC if

$$\left(\sum_{j=0}^{d-1} \binom{n}{j} 3^j \right) \cdot 2^k \leq 2^n$$

Gottesman
Ch. 7

Bound also proves existence of an $[[n, k, d]]$ stabilizer code.

Motivation

Stabilizer codes are nice, but now we need to perform universal computation. Which gates should we apply?

Example: What happens if we apply $RZ(\theta)$ to a ^{single qubit that is part of a} codeword $|\psi\rangle$ in code with stabilizer generators S ?

Stabilized by M' such that $M' RZ(\theta) |\psi\rangle = RZ(\theta) |\psi\rangle$

$$\begin{aligned}
 \textcircled{M'} (RZ(\theta) |\psi\rangle) &= RZ(\theta) |\psi\rangle \quad \leftarrow \begin{array}{l} M \in S \\ M |\psi\rangle = |\psi\rangle \end{array} \\
 &= RZ(\theta) \cdot M |\psi\rangle \\
 &= RZ(\theta) M [RZ^\dagger(\theta) RZ(\theta)] |\psi\rangle \\
 &= \boxed{[RZ(\theta) M RZ^\dagger(\theta)]} RZ(\theta) |\psi\rangle
 \end{aligned}$$

↳ may not still be a Pauli !!

The Clifford group

Better choice: operations that map stabilizer codes directly to other stabilizer codes.

(Source: Gottesman Def. 6.1)

→ my second favourite group

“The Clifford group $\mathcal{C}_n$ on n qubits is the normalizer of $\mathcal{P}_n$ in the unitary group $\mathcal{U}(2^n)$ ”

$$\mathcal{C}_n = \{ U \in \mathcal{U}(2^n) \mid UPU^\dagger \in \mathcal{P}_n \ \forall P \in \mathcal{P}_n \}$$

The Clifford group

Exercise: show that all elements $\mathcal{P}_n$ are in $\mathcal{C}_n$.

$$\underbrace{P Q P^\dagger}_{P, Q \text{ Pauli's}} = \pm \underbrace{Q P P^\dagger}_{= Q} = \pm Q$$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

Exercise: determine the effect of H on $\mathcal{P}_1$.

$$H(\pm X)H = \pm Z$$

$$H(\pm Y)H = \mp Y$$

$$H(\pm Z)H = \pm X$$

$$Y = iXZ$$

$$\begin{aligned} H(\pm Y)H &= H(\pm iXZ)H \\ &= \pm i(HXH)(HZH) \\ &= \pm iZX \\ &= \mp iXZ \\ &= \mp Y \end{aligned}$$

The Clifford group

$$\pm X \quad \pm Y \quad \pm Z$$

Exercise: Determine the size of C_1 .

map $Z \rightarrow$ other Pauli : 6 outcomes

map $X \rightarrow$ other Pauli : 4 outcomes
(that is diff. from
one Z maps to)

24 elements

The size of C_n is

$$|C_n| = 2^{n^2 + 2n} \prod_{j=1}^n (4^j - 1)$$

See derivation [here](#).

The Clifford group

(Source: Gottesman Def. 6.8)

Any element of $\mathcal{C}_n$ can be generated by H_i , S_i , and $CNOT_{ij}$.

$$\mathcal{C}_1: \text{---} \boxed{H} \text{---} \quad \text{---} \boxed{S} \text{---}$$

$$\mathcal{C}_n: \begin{array}{c} j \text{---} \boxed{H} \text{---} j \\ j \text{---} \boxed{S} \text{---} j \end{array} \quad \begin{array}{c} j \text{---} \bullet \text{---} i \\ k \text{---} \oplus \text{---} k \end{array} \quad \forall j, k \in [n]$$

Proof: supplementary exercises (Nielsen and Chuang Ex. 10.40)

Cliffords and binary symplectic representation

Question from the supplementary exercises:

3.4: Determine how Pauli group elements (and thus, stabilizer generators) transform after applying the following unitary operations: H , S , X , Y , Z , and $CNOT$. Compare this with the action of the T gate, and consider the implications on classical simulation of quantum circuits.

4.3: Consider the following four two-qubit Paulis: X_0 , X_1 , Z_0 , Z_1 .

- ① Apply each of the following operations (via conjugation) to these Paulis: H_0 , H_1 , $CNOT$.
- ② Express these Paulis in binary symplectic form, both before and after the operations in part (a). What do you notice about how the matrix representation changes?
- ③ Write down the four stabilizer generators of the 5-qubit code in binary symplectic matrix representation. Based on your observations in part (b), determine the new matrix representation after applying H_2 and $CNOT_{13}$ to each generator.

Cliffords and binary symplectic representation

$$\left(\begin{array}{c|c|c} x_1 & z_1 & r_1 \\ x_2 & z_2 & \\ \hline x_{n+1} & z_{n+1} & \\ \vdots & \vdots & \\ x_n & z_n & r_n \end{array} \right)$$

Clifford circuits can be simulated efficiently on a classical computer.

Main idea: use binary symplectic representation and trace through their action on Pauli generators.

$$HXH = Z$$

$$HYH = -Y$$

$$HZH = X$$

$$SXST = Y$$

$$SYST = -X$$

$$SZST = Z$$

$$X: (1 | 0) \rightarrow (1 | 1)$$

$$Y: (1 | 1) \rightarrow (1 | 0)$$

$$Z: (0 | 1) \rightarrow (0 | 1)$$

add X to Z col
modulo 2

$$\text{BSR: } (1 | 0) \leftrightarrow (0 | 1)$$

$$(\dots 0 \dots | \dots 1 \dots)$$

↓ H

$$(\dots 1 \dots | \dots 0 \dots)$$

switch

X ↔ Z columns

CNOT:

$$ZI \rightarrow ZI$$

$$IZ \rightarrow ZI$$

$$XI \rightarrow XX$$

$$IX \rightarrow IX$$

$$(00 | 10) \rightarrow (00 | 10)$$

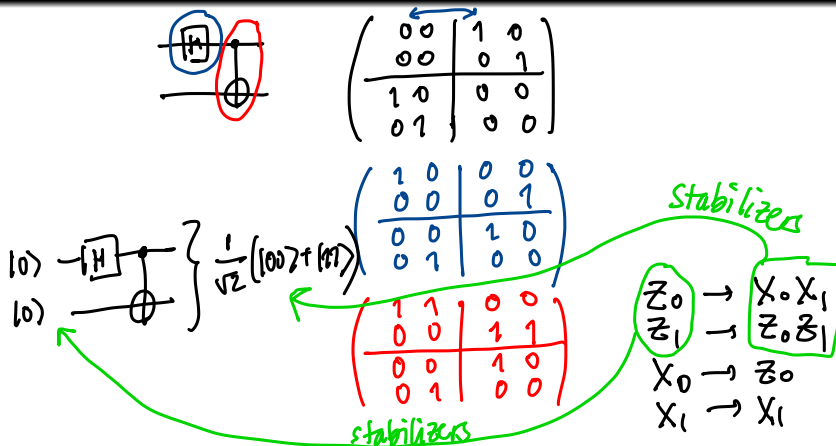
$$(00 | 01) \rightarrow (00 | 11)$$

$$(10 | 00) \rightarrow (11 | 00)$$

$$(01 | 00) \rightarrow (01 | 00)$$

CNOT_{ij}: add col i to j for X
add col j to i for Z

Efficient classical simulation of Cliffords



The Gottesman-Knill theorem

(Source: Gottesman Def. 6.7)

“Given a circuit starting from an initial stabilizer state, followed by a sequence of Clifford group operations and Pauli measurements, which may depend on classical computations performed on previous measurement results, there is an efficient classical simulation of the circuit.”

Application: encoding circuits

Let's derive an encoding circuit for the bit flip code.

Application: encoding circuits

Try yourself: implement the algorithm and determine an encoding circuit for the Steane code.

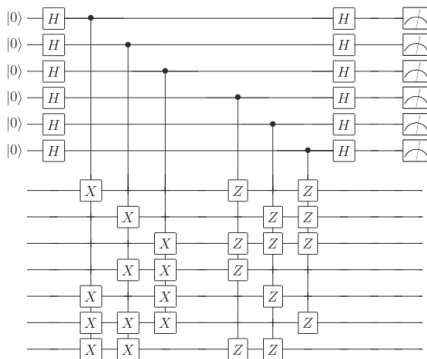


Figure 10.16. Quantum circuit for measuring the generators of the Steane code, to give the error syndrome. The top six qubits are the ancilla used for the measurement, and the bottom seven are the code qubits.

Screenshot: Nielsen & Chuang, chapter 10.5.8.

Next time

Next class:

- Quiz 3
- Start discussing fault-tolerance

Action items:

- 1 Finalize project groups and topics

Recommended reading:

- For this class: Gottesman Ch. 6, N&C 4.5, 10.5
- For next class: Gottesman Ch. 10, N&C 10.6